EX NO -1.1

Design and Draft the given 2D Sketches in modelling software

AIM:

To sketch the given drawing with dimensions using fusion 360 tool

REQUIREMENTS:
1)Autodesk fusion 360
2)Windows 10
3)16 GB of RAM (integrated graphics recommend 2 GB or more)
4)2.5 Mbps or faster download; 500 Kbps or faster upload

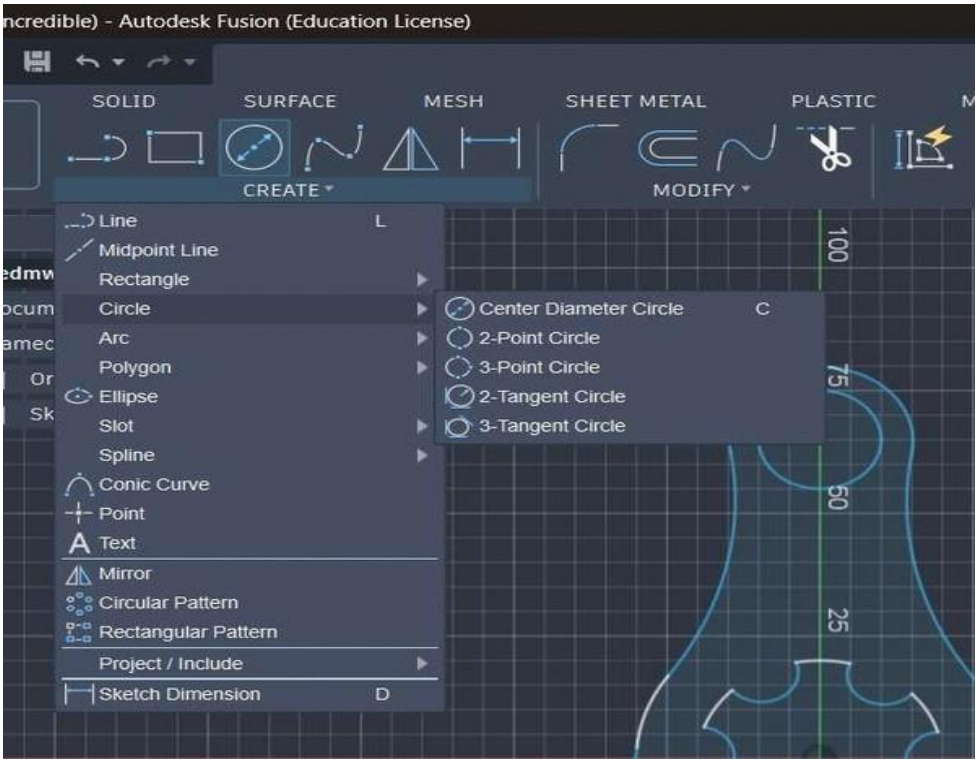
PROCEDURE:

1. Create a New Sketch
- Open Autodesk Fusion 360 and create a new design.
- Select Create Sketch from the Design workspace.
- Choose the XY Plane as the sketch plane.
- Enter the sketching environment and set a suitable view.
- Keep the origin and centre lines visible for reference.

2. Mark the Reference centres
- Take the centre of the main circular feature as the origin.
- Mark the centre of the upper circular feature 65 mm above the origin.
- Mark the centre of the lower circular feature 50 mm below the origin.
- Mark the centres of the left and right side features 26 mm below the origin and 47 mm from the Y-axis.

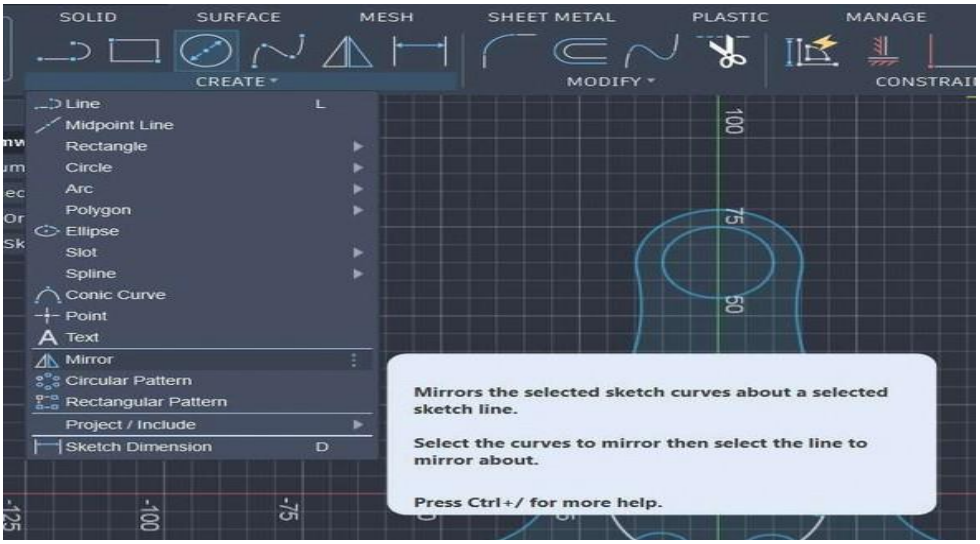
3. Create the Main Circular Features
- At the origin, draw the Ø40 circle

- Create the required R6 circular features inside the Ø40 circle.
- Use the Mirror/Pattern command to create the symmetrical inner features.
- Draw the Ø66 circle with the origin as its centre.
- At the upper centre point, draw the Ø20 and Ø32 circles.
- At the lower centre point, draw the Ø10 and Ø20 circles.
- At both the left and right side centre points, draw the R7/DIA-14 and R10/DIA-20 circles.



4. Complete the Inner Features

- Create the required inner curved features around the Ø40 circle.
- Apply the R6/DIA-12 dimension to the required inner curves.
- Use the Mirror/Pattern command to obtain the remaining symmetrical features.
- Ensure that all inner features are properly aligned with the centre lines.



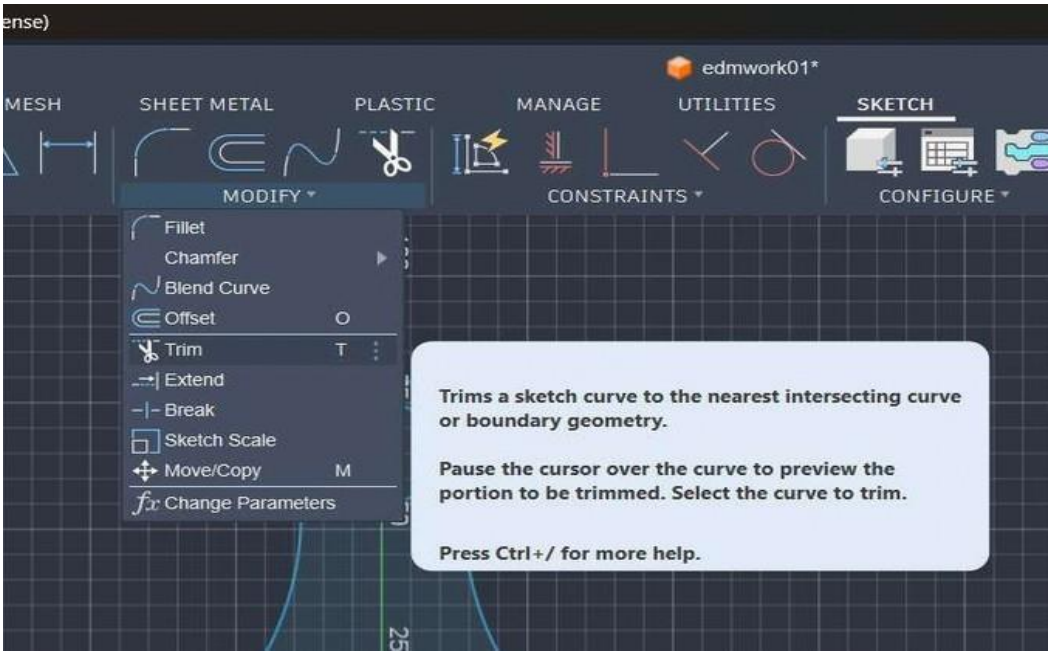
5. Construct the Outer Profile

- Create the required arcs to connect the circular features and form the outer profile.
- Apply the specified R64/DIA-128, R20/DIA-40 and R10/DIA-20 dimensions to the required arcs.
- Use Tangent Constraints to smoothly connect the circles and arcs.
- Use the Mirror command for the required symmetrical geometry.
- Check that the outer profile is smooth, continuous and symmetrical.



6. Trim and Modify

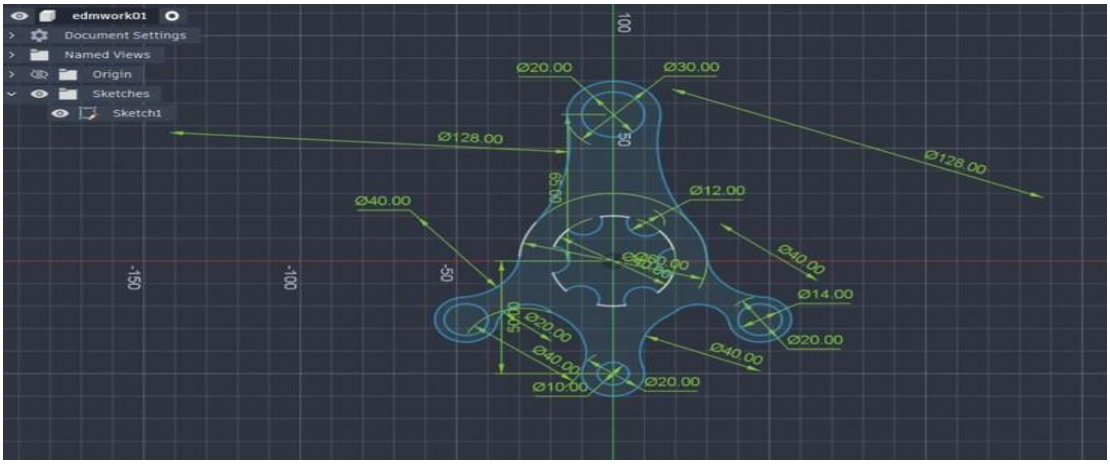
- Select the Trim command under Modify.
- Remove unwanted portions of overlapping circles, lines and arcs.
- Repeat the trimming wherever necessary to obtain the required profile.
- Check that no unwanted or incomplete geometry remains.
- Verify the dimensions and constraints as given.

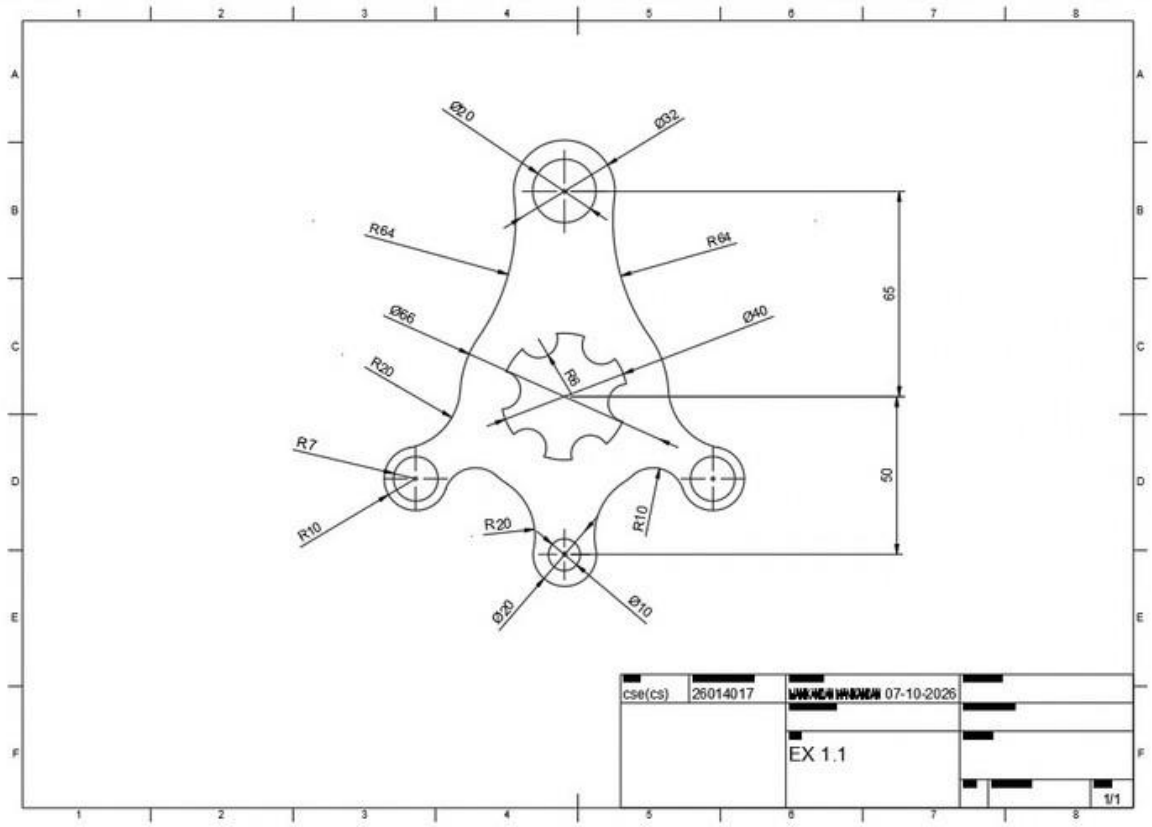


7. Final Verification

- Compare the completed sketch with the given drawing.
- Verify all required dimensions, positions, tangencies and symmetry.
- Ensure that the tangent connections are smooth.
- Check that the complete outer and inner profiles are properly formed.
- Click Finish Sketch to complete the 2D sketch.

OUTPUT:





RESULT:

The given diagram was successfully sketched using Autodesk Fusion 360 with the specified dimensions, geometric features, constraints, and symmetry.